

CONTINUATION — HelixTranslate session-resumption file

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 **RESUME HERE (2026-06-16 ~15:00) — **
**NEZHA HEAVY-TESTING ARC: api-400 fix +
gRPC round-trip + coverage + commentary-escape
fix (all live-validated)**

 **NEZHA HEAVY REAL-SERVICE TESTING — iterations 1→4
(live stack, sink-side, no bluff)**

Heavy testing against the LIVE nezha stack (server-TLS :18443, api :18080, grpc :50061). Live image progression: f3904ccd (pre) → 2d0c925 (api-400 fix) → **ebb82df** (commentary-escape fix; server+grpc+api all on it now, HEALTHY).

- **iter1 5227585** — LLM commentary contamination (trailing meta-prose/ (Note:...)) + create-default-config env-override skip. FIXED + guarded.
- **iter2 2f326cd** — verifier dead-routes 404→honest 503; api /health IDLE false-negative (active gRPC probe); durable --env-file deploy (#3).
- **iter3 (salvage) 6a1aa8c** — api: reject unsupported provider with **400** instead of silent bridge-substitution (§11.4.69/§11.4.120). **RUNTIME RE-VALIDATION DONE (§11.4.108 PENDING→DONE)**: rebooted server on nezha (f3904ccd→2d0c925) and proved sink-side on POST https:// nezha.local:18443/api/v1/translate: RED pre-fix provider:"unsupported-provider"→HTTP **200** silent-sub to llm-novita; GREEN post-fix→HTTP **400** {"error":"unsupported provider: ..."}; VALID deepseek→200 real Serbian Добро јутро, како стаса?; default(empty)→200 Књига лежи на столу. (no regression). Evidence docs/ qa/nezha_revalidate_6a1aa8c_20260616T142656Z/.
- **gRPC round-trip (this run)**: new cmd/grpc-translate-probe (standalone client) drove StartTranslation(/tmp/grpc_in.txt→/tmp/grpc_out.epub, ru→sr, deepseek)→poll GetTranslationStatus running→completed (3m21s)→read output. REAL Serbian Добар дан. Зовем се Иван. Живим у великом граду. ... + valid EPUB (mimetype/container.xml/content.opf/ toc.ncx/chapter1.xhtml) containing the translation. Evidence docs/ qa/ nezha_grpc_roundtrip_20260616T143649Z/.
- **Coverage (this run)**: 5 language pairs (ru↔sr, en→sr, en→ru, ru→en) + script-conversion (sr latin Danas čitajem...) all 200 real; 24-parallel sustained load = 20×200 / 4×429 (clean rate-limit {"error":"Rate limit exceeded"}, p95~1.85s, 2s wall). Evidence docs/ qa/ nezha_coverage_20260616T144123Z/.

- **iter4 e198b17 (this run) — commentary-escape FIX (§11.4.4 test-interrupt → §11.4.102 → §11.4.115 → §11.4.135):** the coverage batch surfaced 1/20 contamination the iter1 fix MISSED — Я пью кофе утром. → 'Ја пијем кафу јутро.\n\n(Using Ekavica dialect and pure Serbian vocabulary as per guidelines)'. FACT root cause: isCommentaryBlock only stripped a trailing ()/[] block when inner started with note or contained translát; this style/dialect aside matched neither. FIX: widened with commentaryParenSignalWords (using/dialect/vocabulary/guideline/register/as per/i used/...); genuine in-content parentheticals preserved (new guard test). Anti-bluff: 2 live-captured RED fixtures FAIL pre-fix/PASS post-fix + KeepsBenignTrailingParenthetical over-strip guard + §1.1 mutation (remove signal-loop→FAIL→restore→PASS); pkg/translator/llm+pkg/translator GREEN; vet clean. Deployed (server+grpc+api→ebb82df), re-validated live: 18/18 clean (the exact pre-fix-contaminating source now CLEAN), task-1 400-fix unregressed. Evidence docs/qa/nezha_revalidate_e198b17_20260616T145411Z/.

◆ OPERATOR REVIEW QUEUE (heavy-testing arc)

1. **.env.nezha line 48 malformed** (§11.4.10 — value NEVER printed/committed): python-dotenv could not parse statement starting at line 48 warns on every nezha deploy ⇒ OPENAI/ANTHROPIC keys may not be injected via that path (note: env NAMES OPENAI_API_KEY/ANTHROPIC_API_KEY ARE present <set> in containers, so injection is partial/via another mechanism). **Action:** operator reformats .env.nezha to one NAME=value per line (no continuation/quotes spanning lines around line 48).
2. **Live LLMsVerifier upstream not cross-network wired** ⇒ /api/v1/verified-models cannot serve real upstream data; bridge model-selection at server startup blocks ~3.5min on bridge.Open (5m30s timeout) before binding (NOT a crash — it binds + HEALTHY). **Action:** enable the nezha LLMsVerifier upstream wiring + set its APIURL so /api/v1/verified-models returns real data and startup model-selection resolves fast.
3. **scripts/nezha-deploy.sh reboot dependency-ordering bug:** rebooting a service that has dependents (e.g. grpc-server, which api+server depend_on) fails podman rm ("has dependent containers") and then compose up collides on existing names ⇒ the dependent service silently keeps the OLD image (§11.4.108 risk — caught this run, recovered by stop+rm dependents-first then up). **Action:** make reboot stop+rm the dependent subtree before rm'ing a depended-on service (or use podman rm --depend/up --force-recreate once supported).

 (prior) RESUME HERE (2026-06-16 ~15:35) —
 **NEZHA DISTRIBUTED BOOT COMPLETE**
(full stack live + real translation verified) · §11.4.153
video program 43 confirmed · release 2.3.0 published

 **NEZHA DISTRIBUTED BOOT — DONE + SINK-SIDE
VERIFIED (operator-CRITICAL task, COMPLETE)**

The whole HelixTranslate System is DEPLOYED + RUNNING on **nezha.local** (Linux 6.12, 8c/62GB, podman 5.7.1, SSH milosvasic keyed) via the **containers submodule** (§11.4.76: launcher containers/cmd/deploy-stack commit 20173e8, parent gitlink 6296c03; Phase A defs 2166d2c; JWT fix 1b29668). **All 6 services Up HEALTHY**: helixtranslate-
{postgres,redis,monitor[:18090],grpc[:50061],api[:18080],server-TLS[:18443]}
(ports remapped to avoid host conflicts). **REAL translation round-trip
VERIFIED sink-side (§11.4.69)**: POST https://nezha.local:18443/api/v1/
translate en→es → HTTP 200, provider llm-novita → "El viejo faro se alzaba
solitario en la rocosa orilla." Secrets in gitignored .env.nezha on nezha (real
JWT_SECRET + all provider keys, chmod-safe, NEVER committed §11.4.10).
Stack left RUNNING for heavy testing. NEXT: run the heavy/real-production-
service test suites against the live nezha endpoints; any bug → systematic-
debugging → fix → rebuild → reboot → re-validate (NO bluff).  I premature-
stopped progressing agents TWICE during the boot — remote ops are 30-60+min;
BE PATIENT, trust completion, suspect hang ONLY if no completion + no ssh/
podman/curl subprocess + zero nezha state-change after 60+min.

§11.4.127 SHORT resumption (paste this first sentence into a fresh session)

Read docs/CONTINUATION.md (this file) + .remember/remember.md FIRST, then
git fetch --all --prune; **NEZHA BOOT COMPLETE** (full stack live on
nezha.local via containers submodule, real translation verified — see green block
above; run heavy testing against it next); the no-local-runtime release
helix_translate-2.3.0 is PUBLISHED (tag on main + 10 owned submodules;
helix_qa tag OWED §11.4.119); the **§11.4.153 per-feature videoconfirmation
program** continues — docs/features/Status.md Rev 14, **43/496 features
video-confirmed** (waves 1+2+3a: bridge-ensemble/catalogue, DOCX/PDF/gRPC/
HTML-MD, providers novita/mistral/groq/siliconflow/cerebras/sambanova — real
bridge responses, ffmpeg+content-verified, NO bluff), model source =
LLMsVerifier bridge (real keys present), HelixAgent incorporated (helix_agent/
pinned) NOT integrated (42-dep graph = deferred operator decision). **A release-
critical default-path defect the video program CAUGHT is FIXED + re-
reviewed GO + guarded: commit 7c2118a** (VerifiedFactory routes verifier-
selected models through the whitelist-immune path whenever a key exists from
ANY source; env-independent §11.4.138 guard; proven GREEN with all 36
*_API_KEY unset).  **RECORDING ROOT-CAUSE (corrected via systematic-
debugging — do NOT repeat the earlier thrash)**: recording is INTRINSICALLY
SLOW (~5–6 min/video: real LLM call + asciinema→agg→ffmpeg + ffmpeg +
frame-analysis + agent reasoning). The bridge is NOT slow/hung (best-model=1s
cached; 600s provider timeout is intentional for long translations). Long stale-output
gaps are NORMAL — do NOT misread them as "hung" and premature-stop (that

churn killed progressing agents, e.g. wave-3a was stopped AFTER it had committed). DISCIPLINE: dispatch small recording units (2–4 videos), be PATIENT (give 25–30 min before even suspecting a hang), check the DECISIVE signal (did it commit / are mp4s landing / is there an asciinema/ffmpeg/translate subprocess) before stopping. Full autonomous coverage of ~90 remaining eligible features ≈ many hours — a long steady iterative program, not a sprint. Continue the loop (§11.4.126), no force-push (§11.4.113), no git add -A (§11.4.30), never touch helix_qa (§11.4.119)/helix_agent build; SINGLE main-checkout committer at a time (§11.4.84 — reviews read-only, recorders worktree-no-commit); contested checkout (helix_code session shares machine) — verify branch/lock/sync before every commit. NEXT: wave-3 owed (providers deepseek/cerebras/sambanova/hyperbolic/gemini/hipu/cohere/fireworks/openrouter + REST endpoints + verify-models + workable-items), then continue toward full coverage; web/desktop = honest operator-attended (§11.4.3/§11.4.52). Tracked: thread CLI -max-tokens through bridgeTranslator; commit untracked docs/design/LLMSVERIFIER_BRIDGE.md/.html; clean stale worktree .claude/worktrees/agent-a8ea4f6230cb2ee40.

§11.4.127 FULL resumption (live-state anchors — verified this session against git)

- **Parent HEAD 3ce413e** on main, local == origin/main (verified git fetch --all this session; pushed FF to both upstreams github+HelixDevelopment). Untracked build/* + root binaries are EXTERNAL build-artifact churn (a gemini-cli/data-agent session rebuilds them) — NOT committed (§11.4.30).
- **⚠️ CONTESTED CHECKOUT (§11.4.84/§11.4.119):** an EXTERNAL gemini-cli/data-agent process intermittently runs git checkouts + has left a stale .git/index.lock in this repo. BEFORE EVERY commit: (a) confirm git symbolic-ref -q HEAD == refs/heads/main, (b) ls .git/index.lock absent (remove a STALE lock only if no live git process), (c) git fetch --all + confirm local == origin/main. Every R-5 commit survived via this protocol.
- **R-5 commits this session (all FF-pushed):** 102c773 CM-NO-LOCAL-RUNTIME gate + paired meta-test · abc0912 Part-B NITs (bridge/format test asserts + bridgeOpener seam + detector XHTML anchor) · 7036a30 config.worker.json default_provider ollama→openai · 3ce413e §11.4.153 ledger 39 rows Obsolete + verifier-doc framing fix.
- **Prior phase-2 commits (P-1..R-4):** 8325f9d R-4 remove worker configs + SSH/llama/ollama scripts · 86175d0/earlier R-4 remove SSH binaries + pkg/sshworker + pkg/modelsbridge + llamacpp_provider + markdown bridge wiring; P-1 BestClient e5307ce.

Bridge phase-2 — the CRITICAL no-local-runtime mandate (state: NEAR-COMPLETE)

Operator-CRITICAL: NO local runtime — only LLMsVerifier strongest models via pkg/bridge, bridged to components AND this Claude Code session. Plan: **docs/design/BRIDGE_PHASE2_EXECUTION_PLAN.md** (+ design docs/design/LLMSVERIFIER_BRIDGE.md, removal map docs/design/LOCAL_RUNTIME_REMOVAL.md).

LOCKED operator decisions: D1 remove local runtimes entirely; D2 forbid ALL local runtimes (llama.cpp + Ollama); R1 → **R-4 decision (2026-06-16): keep the distributed/API path, remove SSH-local** (the SSH-local binaries + worker body Obsolete §11.4.90; version_manager/ssh_pool/distributed coordination KEPT); R2 **require API keys everywhere** (no offline/local fallback — honest error/SKIP, never a silent local fallback); D4/D5 **top-1 + fallback, in-process; ensemble = "Provider-diverse verified"**.

Status — DONE:

- **P-1 / R-1** BestClient/BestTranslator + all default-path components redirected to the bridge.
- **R-2** Ollama removed · **R-3** llama.cpp removed · **R-4** SSH-local binaries + pkg/sshworker + pkg/modelsbridge + local worker configs/scripts removed (operator-confirmed).
- **R-5 (this session):** CM-N0-LOCAL-RUNTIME pre-build gate (3 arms, default-path-only) + paired §1.1 meta_test_no_local_runtime.sh (baseline PASS / Mut1 Arm1 FAIL / Mut2 Arm3 FAIL / Neg PASS); Part-B NITs; config.worker.json fixed; §11.4.153 ledger 39 rows Obsolete (docs_chain features re-synced HTML/PDF/DOCX); AGENTS.md + README.md + this CONTINUATION updated.
- **§11.4.40 FULL RETEST: PASS** (authoritative main-checkout, HEAD f9c52a2; evidence docs/qa/bridge_phase2_retest_20260616/SUMMARY.md, commit d1bed61): sweep all-9-gates green · build clean · functional 55 pkgs OK · -race only pre-existing test/distributed (§11.4.7) · meta-tests all bite (incl. meta_test_no_local_runtime.sh) · determinism stable · product challenges pass. Two prior worktree-retest FAILs proven ENVIRONMENT artifacts (worktree missing submodules; worktree checkout mtime) — same gate PASSES on the real main checkout; NOT regressions.

PHASE-2 CLOSED + RELEASED. 🟡 **helix_translate-2.3.0** (operator "do tag now" 2026-06-16; safest zero-churn calls: version = current VERSION 2.3.0 no bump; set = main + 10 owned submodules, helix_qa excluded §11.4.119). Annotated tag created + pushed to ALL remotes across **main (6b81a10) + containers/challenges/doc_processor/llm_orchestrator/llm_provider/vision_engine/llms_verifier/constitution/docs_chain/security** (each at its gitlink-pinned commit), verified via ls-remote (main ^{ }→6b81a10). **OWED:** helix_qa tag (off-limits to this session §11.4.119 — operator/its session to apply helix_translate-2.3.0).

Non-blocking follow-ups (not product defects): (a) host_no_auto_suspend challenge FAILs = HOST not hardened against auto-suspend — environment (§11.4.3), host power config NOT changed autonomously (CONST-033), operator action if desired; (b) internal/working/config.distributed.{json,test,thinker}.json legacy llamacpp/gguf tokens (KEPT distributed path) — future R-cycle if distributed worker API-migrated; (c) codebase_hash_report.json generator needs a submodule-exclusion before faithful refresh (not a gate dep); (d) format-detector XHTML-branch advisory (conservative, zero real-tree impact) — optional future hardening.

Standing constraints (binding every session)

- **No force-push EVER** (§11.4.113 — merge-onto-latest-main, FF-only); **never git add -A** (§11.4.30); **never touch the helix_qa submodule** — another session owns it (§11.4.119 single-resource-owner); **never touch owned submodules** in this R-5 scope.
 - **No host power-management** (§12 / CONST-033). **Require API keys, NO local llama.cpp/Ollama/SSH-distributed runtimes.** Operator-blocked → AskUserQuestion (§11.4.66). Endless autonomous loop is the DEFAULT (§11.4.126).
 - **Working API keys:** DEEPSEEK_API_KEY + GROQ_API_KEY present (real translations work); GEMINI/ZHIPU invalid/quota-blocked (honest errors). NEVER print key values (§11.4.10).
 - **Contested checkout:** apply the pre-commit vigilance protocol above before every commit (§11.4.84/§11.4.119).
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🚫 (SUPERSEDED 2026-06-15 ~22:30 — subagent runtime has since reset; reconciliation + P-1 landed) RATE-LIMIT BLOCK + OWED CRASHED-STREAM WORK (2026-06-15 ~19:05)

The subagent runtime hit a session/usage limit (subagents die instantly, ~178 tokens). NO new subagents can run until the **7:10pm Europe/Moscow reset**. Two phase-1 bridge streams CRASHED mid-work (§11.4.147 crash≠done — work OWED, partial state PRESERVED uncommitted in the worktree, do NOT discard, do NOT commit-incomplete §11.4.84/§11.4.121):

LLMsVerifier-bridge mandate (operator CRITICAL 2026-06-15): no local llama.cpp — only LLMsVerifier strongest models + bridge to components AND this Claude Code session. Design: docs/design/LLMSVERIFIER_BRIDGE.md; removal map: docs/design/LOCAL_RUNTIME_REMOVAL.md. LOCKED operator decisions: D1 remove llama.cpp entirely; D2 **forbid ALL local runtimes** (llama.cpp + Ollama + SSH/distributed); D3 agent access = **CLI + MCP**; D4/D5 **top-1 + fallback, in-process** pipeline (env *_API_KEY keys, no service); R1 **Obsolete** cmd/translate-ssh + cmd/ssh-translation + llamacpp/ollama worker configs (§11.4.90); R2 **require keys everywhere** (no offline path — tests FAIL not skip when keys absent).

OWED CRASHED STREAMS (respawn after reset, §11.4.147 + §11.4.84 quiescence-check first):

- **bridge-core** (was a9659f5) — INCOMPLETE, does NOT build: cmd/model-bridge/main.go:52: undefined: runMCP (crashed before writing the MCP stdio server). Preserved partial: pkg/bridge/ (BestTranslator/Invoke/BestModel/ListVerified facade), cmd/model-bridge/main.go (CLI), internal/verifier/provider_resolver.go+_test.go (numeric-ProviderID gap-fix). RESPAWN to: finish runMCP (MCP stdio server) + .mcp.json wiring + full tests + real best-model/invoke evidence + docs/guides/LLMSVERIFIER_BRIDGE_USAGE.md; then §11.4.142 review + commit.
- **web-UI** (was aaf872a) — BUILDS clean (pkg/api + cmd/server). Preserved: pkg/api/dashboard.go+dashboard_test.go, pkg/api/handler.go (M) —

wires the unrouted Web Dashboard translation UI (the real §11.4.153 web gap).
OWED: independent §11.4.142 review + real-server real-translation evidence
run, then commit. (Note: a later phase swaps its provider to
bridge.BestTranslator.)

NEXT (post-reset, phased per LOCAL_RUNTIME_REMOVAL.md R-0→R-5): (1)
respawn bridge-core to finish; (2) review+commit web-UI; (3) phase-2: redirect all 9
components to bridge.BestTranslator, remove Ollama, remove llama.cpp,
Obsolete SSH/distributed (R1), add CM-N0-LOCAL-RUNTIME gate + paired mutation,
require-keys-everywhere (R2); (4) HelixQA web-runner browser+video backend
extension (needs the helix_qa session) → then the real web-dashboard video. Web-
dashboard video remains PENDING until both the UI wiring lands AND HelixQA
gains a web/video backend.

DONE + PUSHED this session (all remotes, FF-only): constitution §11.4.153
(74e7fbe, 10 remotes); parent
58178a5→a5e8866→86bf81e→d53e085→fb74c2d→87cd2be→e73e0b7→fb265e7;
submodules llm_provider 724e284, containers 2a9923a, llms_verifier 89719e2b.
§11.4.153 mandate + 470/472 ledger (21 video-confirmed) + DOCX & PDF output
writers + multipass wiring + cli title/atomic-write fixes + prep-translator/server/
REST bug fixes + README §11.4.57 + all 9 submodules GREEN.

✓ §11.4.153 LEDGER RE-SYNC (2026-06-15, +DOCX output + -multipass video + cli atomic- write)

Three capabilities became real+evidenced this session and landed+pushed; the
feature ledger (docs/features/Status.md + Status_Summary.md) is re-synced
honestly (docs UNCOMMITTED — conductor commits, no git add -A):

- **DOCX output** — gap → **Implemented** (pure-Go OOXML writer pkg/ebook/
docx_writer.go, commit 87cd2be). Verified this session by re-producing a
.docx via the shipped writer → file = Microsoft Word 2007+, all 4
OOXML parts, real translated content, correct XML escaping. **Real-artifact-
confirmed (NOT video — no .mp4 yet, video PENDING).**
- **-multipass multi-pass verification** — now **video-confirmed**:
helixtranslate-cli-multipass-verify-20260615.mp4 (ffprobe 888x630,
93 frames, 9.3s) shows "Step 4: Multi-pass Polishing ✓ Success — Polished
over 1 pass with deepseek" + real Spanish output. Engine wired into unified-
translator via -multipass in d53e085.
- **cmd/cli atomic-write / no-clobber** (87cd2be) — write-safety fix, guard cmd/
cli/no_partial_output_test.go (unit-proven, no video required); title
path-leak fixed in d53e085.
- **Honest counts**: video-confirmed **20** → **21** (the -multipass flip) + **1 real-
artifact-confirmed** (DOCX); Status/Status_Summary Rev 2→3; not-
implemented gap rows 2→1.
- **Exports re-synced**: docs_chain sync features (6 derived nodes) →
docs_chain verify features = **in-sync (RC=0)**; HTML+PDF+DOCX
siblings regenerated for both docs (all file-verified).
- **Remaining gaps now**: **PDF output** (no writer); cmd/translator local path
STUB; llm_provider gofmt hygiene (23 flagged files); cosmetic docx title-

twice; web-dashboard translation video **OPERATOR-BLOCKED** (HelixQA-gated). translate-ssh/ebook-translator stay **OPERATOR-BLOCKED**.

✓ **GAP-CLOSURE BATCH (2026-06-15, video/coverage-surfaced gaps — all reviewed GO + pushed)**

Three gap-closures (each RED-first §11.4.115, mutation-verified, independent batch review VERDICT GO §11.4.142) landed + pushed:

- **Parent d53e085** (pushed all remotes): (1) **cmd/cli title path-leak FIXED** — TXT parser set `Metadata.Title=filepath` → translated → leaked into output EPUB `<dc:title>`; `sanitizeBookTitle()` added at the cmd/cli input→book seam (genuine titles pass through; real run `<dc:title>Tiha bašta</dc:title>`, NO LEAK). (2) **pkg/verification multipass engine WIRED** (§11.4.124 — git-history proved it was NEVER imported by any cmd/, born dormant 717bf18, finished real code not scaffolding) via opt-in `-multipass` flag (default OFF, back-compat; guardrails: `passes≥1`, base preserved on error/empty); RED→GREEN httpstest + real DeepSeek "Step 4: Multi-pass Polishing ✓"; §11.4.150 research docs/research/20260615_video_surfaced_fixes/verification_multipass_wiring.md.
- **llm_provider 724e284** (pushed LLMProvider remote; parent gitlink bumped in d53e085): contract test for the interface-only pkg/provider (zero-coverage §11.4.28 gap; mutation-proven; suite green).

NEXT (open gaps, tracked — none blocking the current scope): implement DOCX/PDF **output** writers in pkg/ebook (input-only today); implement cmd/translator local path (STUB "not yet implemented"); fix cmd/cli silent-fail-exit-0 when `-model` omitted (incidental finding); record a `-multipass` video for §11.4.153 ledger confirmation (currently real-run+RED→GREEN evidenced, no .mp4 yet); gofmt the 23 pre-existing flagged files in llm_provider; web-dashboard translation video is **OPERATOR-BLOCKED** (HelixQA-gated, separate session). translate-ssh/ebook-translator stay **OPERATOR-BLOCKED** (need remote SSH/llama.cpp host).

🎬 **§11.4.153 FEATURE-STATUS VIDEO-CONFIRMATION SYNC (2026-06-15, ~25 real videos → ledger honest)**

The §11.4.153 per-feature Status ledger is now in honest sync with the real video evidence captured this session (docs uncommitted — conductor to commit; do NOT git add -A).

What landed before this sync (already pushed):

- **§11.4.153 doc set** (docs/features/Status.md + Status_Summary.md, 470/472 features, four-format HTML+PDF+DOCX) shipped in 58178a5; constitution bump 74e7fbc.
- **3 video-surfaced bugs FIXED + pushed** in a5e8866 (the green suite had missed all 3, §11.4.4/§11.4.138): (1) preparation-translator DEAD — DeepSeek

API key never reached config (-api-key/env not wired) → "DeepSeek API key is required", no analysis JSON; (2) cmd/server TLS startup — config with no tls_cert_file failed to start; fix backfills self-signed certs in LoadConfig; (3) REST POST /api/v1/translate ignored target_lang (hardcoded) → always Serbian regardless of requested language.

- ~25 real videos captured under /Volumes/T7/Downloads/Recordings/helixtranslate-*.mp4 (all ffmpeg-verified dur>0+frames>0; key ones content-verified frame-by-frame this session).
- All 9 owned submodules GREEN — coverage logs qa-results/submodule_coverage_20260615/.
- README §11.4.57 Tracked-Items doc-link section added (prior session).

This sync (Status.md/Status_Summary.md Rev 1→2 + exports re-synced):

- **Video-confirmed: 1 → 20** feature rows (≈4.3% of 472), each flipped to Confirmed: ONLY after the cited .mp4 was content-verified (§11.4.6 no overclaim). Confirmed: unified-translator (DeepSeek translate; EPUB→TXT / HTML→EPUB / FB2→EPUB / FB2→FB2; Serbian-Cyrillic; -script Cyrillic→Latin; format-detection; help/version; -verify), REST API (POST /api/v1/translate real DeepSeek + target_lang FIXED, TLS startup FIXED, /health/version/providers), gRPC (real EN→ES), markdown-translator (EPUB→MD round-trip), preparation-translator (FIXED), cmd/cli (real DeepSeek), monitor-server (live WS hub), DeepSeek library client.
- **Honest gaps recorded (§11.4.6, NOT confirmed):** DOCX output + PDF output UNIMPLEMENTED (no writer in pkg/ebook; unified-translator rejects them; 2 new gap rows added → 472 total); -verify ≠ pkg/verification multipass engine (engine has no CLI entry point wired in); cmd/translator local path = STUB ("local translation not yet implemented"); cmd/translate-ssh + cmd/ebook-translator = OPERATOR-BLOCKED (need remote SSH/llama.cpp worker host); GET /api/v1/verified-models = 404 (LLMsVerifier disabled in config); cmd/cli minor cosmetic file-path-leak into title metadata.
- **Exports + fingerprint re-synced** via docs_chain sync features (applied: 6 derived nodes — HTML/PDF/DOCX × 2 docs) → docs_chain verify features = in-sync (RC=0); evidence qa-results/docs_chain/20260615T144104Z/sync.log; docs_chain state.json = §11.4.86 content-hash fingerprint store.

NEXT (open gaps queued): implement DOCX/PDF output writers in pkg/ebook; wire pkg/verification multipass engine into a CLI entry point; implement cmd/translator local path; produce videos for remaining PENDING runtime features (web dashboard translation, other providers) to lift coverage. translate-ssh/ebook-translator stay OPERATOR-BLOCKED until a remote SSH/llama.cpp host is provided.



WAVE 15 — SECOND-PASS + determinism (campaign 101→110; second pass STILL productive)

Parent HEAD 6c89910, build/vet/test green, git OK, llms_verifier pointer intact.

- **pkg/ebook 6c11dee (#102):** EPUB parser set chapter.Title to the zip entry name → EPUB→translate→EPUB rendered "OEBPS/chapter1.xhtml" as every chapter heading + TOC entry (core use case). Fix = extractChapterTitle (